FX Time Range Breakout
Historical profitability and parameter consistency

Research question
Do JPY pairs show broader and more locally consistent historical breakout profitability than non-JPY pairs?
Original dataset
28 FX pairs: 7 JPY and 21 non-JPY.
Test date inputs: 1 May 2018–31 August 2026.
Broker server: CapitalPointTrading-MT5-4. Modelling: 1-minute OHLC.
318,136 optimization rows; 286,832 eligible configurations.
Experiment
The original 28-pair study fixes the stop loss at 100% of the range, placing it at the opposite boundary. Range starts, durations and closing times vary in 30-minute steps; both breakout modes are included. All scheduled times use the broker clock.
Additional USDJPY scans use five-minute range-start and duration steps in two-signal mode, with exits at 17:00, 20:00 and 23:00. These use the same dates and 1-minute OHLC modelling and are analysed separately from the original dataset.
Tools
Excel inspection → Python parsing and visualization → MySQL analysis → Power BI and DAX reporting.
Limitations
Each row represents an alternative configuration tested over the historical period. Configurations overlap and pairs share currencies. These exploratory results require real-tick and out-of-sample validation.

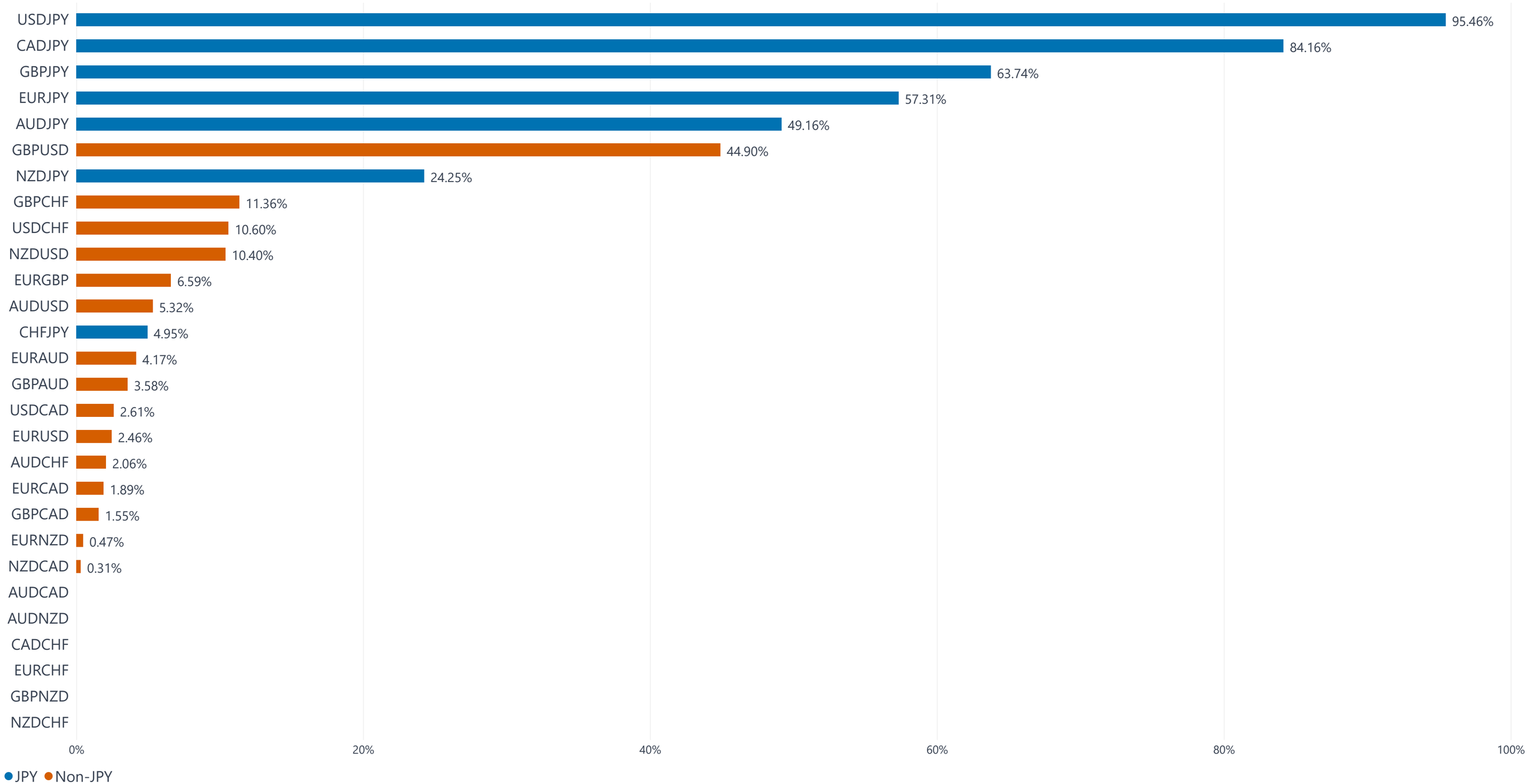
Metric definitions

Eligible configuration
At least 120 scheduled minutes between range end and closing time, with at least one trade.
Profitable-setting share
Percentage of eligible configurations with positive net profit. This measures coverage across settings, rather than the percentage of winning trades.
Recovery factor (RF)
Net profit divided by maximum balance drawdown, both measured in money.
Profit factor (PF)
Total profits from winning trades divided by the absolute total losses from losing trades. A value above 1 indicates historical profitability.
Custom correlation R
Pearson correlation between trade number and cumulative net profit normalized per lot. Values near +1 indicate a nearly linear upward path in trade order.
Neighbourhood support
A configuration is supported when it and all six neighbouring configurations are profitable. Each neighbour changes one timing input by ±30 minutes, keeping the other inputs fixed.

Raw Backtests	Eligible Backtests	pair_code	Profitable Backtests	Profitable Settings %	Median RF (Profitable)	Average PF (Profitable)	Average Equity DD % (Profitable)
11,362	10,244	USDJPY	9779	95.46%	2.9025	1.1287	0.9051%
11,362	10,244	CADJPY	8621	84.16%	2.3030	1.1032	0.8795%
11,362	10,244	GBPJPY	6530	63.74%	1.1578	1.0529	1.3712%
11,362	10,244	EURJPY	5871	57.31%	0.6743	1.0408	1.3030%
11,362	10,244	AUDJPY	5036	49.16%	0.5705	1.0415	1.0751%
11,362	10,244	GBPUSD	4600	44.90%	0.6698	1.0383	1.8445%
11,362	10,244	NZDJPY	2484	24.25%	0.4120	1.0311	1.1880%
11,362	10,244	GBPCHF	1164	11.36%	0.2589	1.0215	2.1414%
11,362	10,244	USDCHF	1086	10.60%	0.1712	1.0140	1.4323%
11,362	10,244	NZDUSD	1065	10.40%	0.2658	1.0211	1.2082%
11,362	10,244	EURGBP	675	6.59%	0.1943	1.0185	1.3332%
11,362	10,244	AUDUSD	545	5.32%	0.2358	1.0185	1.0579%
11,362	10,244	CHFJPY	507	4.95%	0.1182	1.0105	1.9335%
11,362	10,244	EURAUD	427	4.17%	0.1996	1.0148	1.3718%
11,362	10,244	GBPAUD	367	3.58%	0.1164	1.0092	1.8628%
11,362	10,244	USDCAD	267	2.61%	0.2250	1.0197	0.6834%
11,362	10,244	EURUSD	252	2.46%	0.2332	1.0141	1.1083%
11,362	10,244	AUDCHF	211	2.06%	0.1523	1.0142	1.0766%
11,362	10,244	EURCAD	194	1.89%	0.3611	1.0192	1.0399%
11,362	10,244	GBPCAD	159	1.55%	0.1566	1.0131	1.8166%
11,362	10,244	EURNZD	48	0.47%	0.0556	1.0059	2.0997%
11,362	10,244	NZDCAD	32	0.31%	0.1086	1.0087	0.9283%
11,362	10,244	AUDCAD	0	0.00%			
11,362	10,244	AUDNZD	0	0.00%			
11,362	10,244	CADCHF	0	0.00%			
11,362	10,244	EURCHF	0	0.00%			
11,362	10,244	GBPNZD	0	0.00%			
11,362	10,244	NZDCHF	0	0.00%			
318,136	**286,832**		**49920**	**17.40%**	**1.0292**	**1.0664**	**1.2071%**

Profitable Settings by FX pair

Share of eligible settings | SL=100% of range | 1-minute OHLC

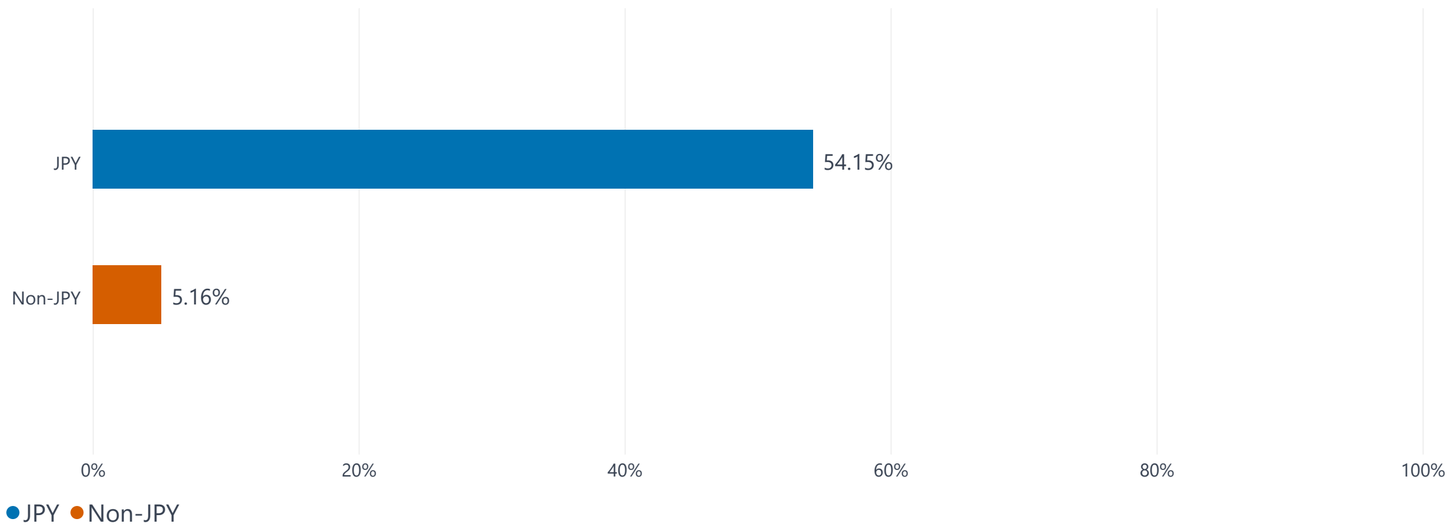


JPY versus non-JPY pairs

Pair Group	Mean Pair Profitable Share %	Pair Count	Median Pair Profitable Share %
JPY	54.15%	7	57.31%
Non-JPY	5.16%	21	2.06%
Total	**17.40%**	**28**	**3.88%**

Mean profitable-setting share by group

Equal weight per pair | 7 JPY pairs and 21 Non-JPY pairs



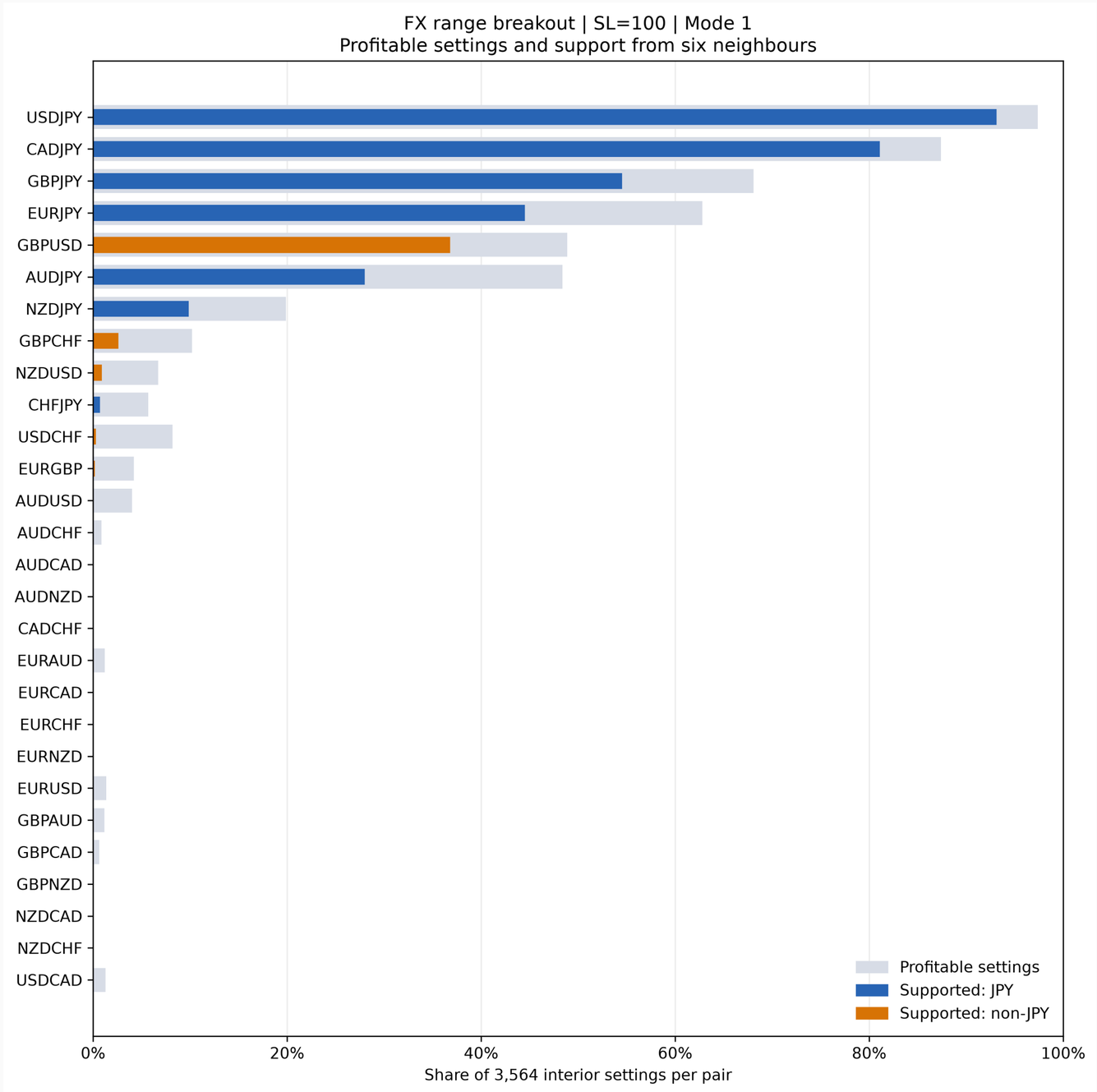
JPY pairs show broader historical profitability

The mean profitable-setting share is 54.15% for JPY pairs versus 5.16% for non-JPY pairs—a gap of 48.99 percentage points. Each pair receives equal weight. These percentages describe profitable backtest configurations. The configurations overlap and currency pairs share currencies, so they are not independent observations. Real-tick and out-of-sample validation remain outstanding.

Profitable-setting coverage by currency group

Currency group	Pairs	Mean profitable-setting share	Median profitable-setting share
JPY	7	54.15%	57.31%
USD	7	24.54%	10.40%
GBP	7	18.82%	6.59%
CAD	7	12.93%	1.55%
EUR	7	10.41%	2.46%
AUD	7	9.18%	3.58%
NZD	7	5.06%	0.31%
CHF	7	4.14%	2.06%

JPY-linked pairs have the highest mean (54.15%) and median (57.31%) profitable-setting coverage in this historical sample. Each currency group contains seven pairs, with equal weight per pair. Each pair belongs to two currency groups, so the groups overlap. Every non-JPY currency group also includes its JPY cross. Coverage includes both breakout modes and 10,244 eligible configurations per pair.



Local parameter consistency

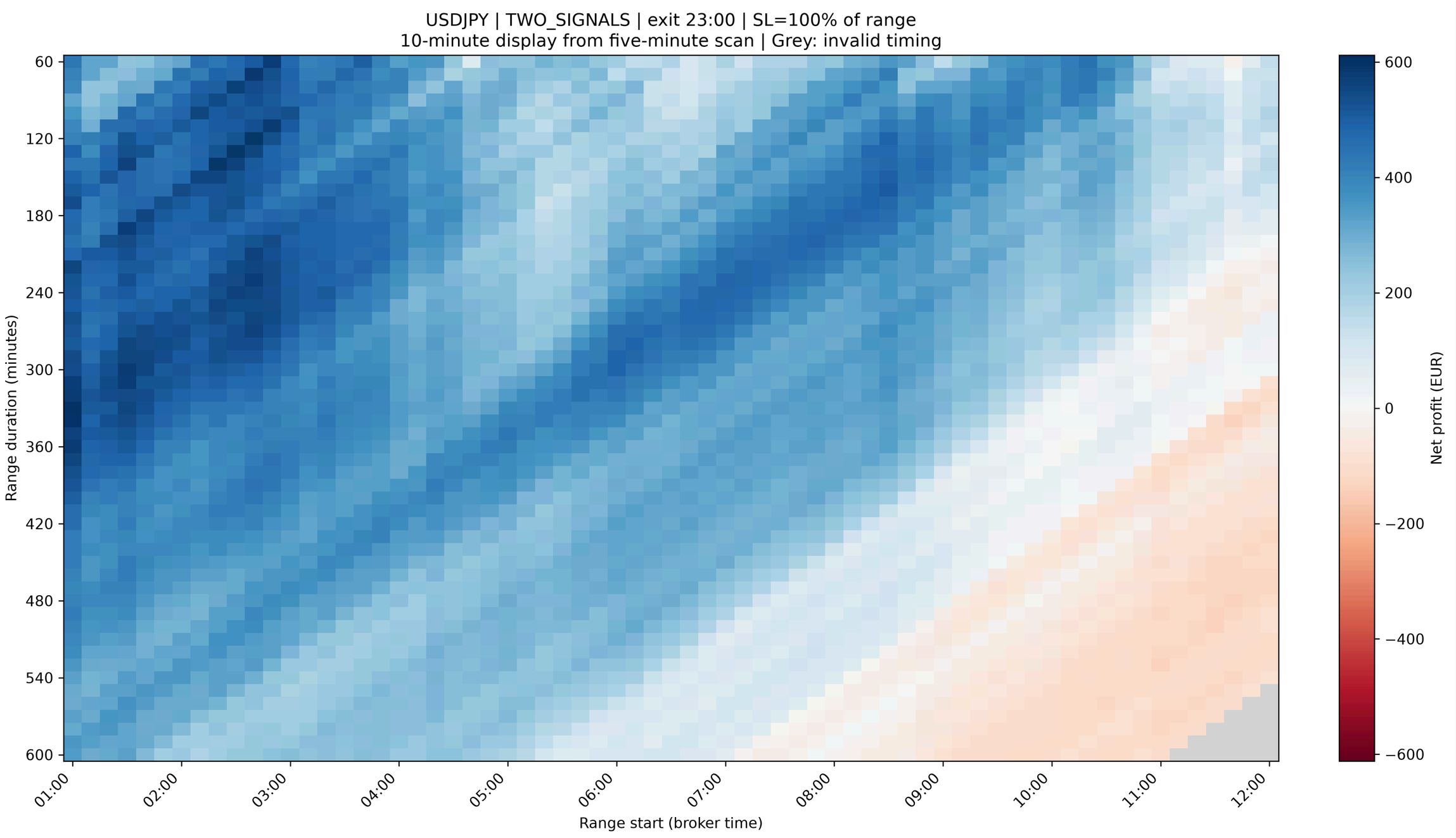
A setting is supported when it and all six neighbours are profitable. Each neighbour changes one timing input by ±30 minutes.

Scope: two-signal mode, 1-minute OHLC, 3,564 interior settings per pair.

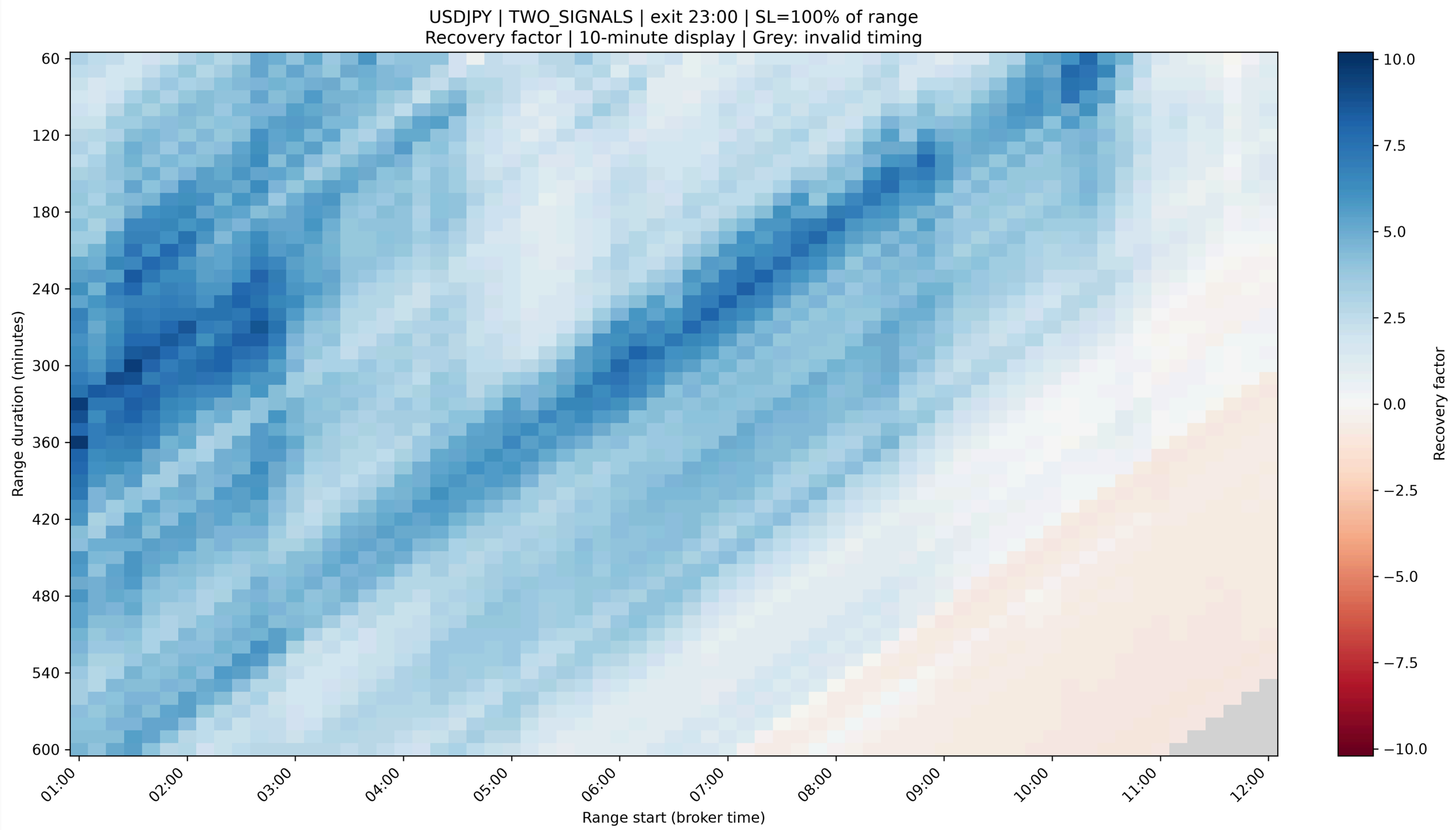
Mean supported share: **44.55% for JPY pairs**, versus **1.95% for non-JPY pairs**.

Excluding USDJPY and CADJPY leaves a JPY mean of **27.53%**; the non-JPY mean remains **1.95%**.

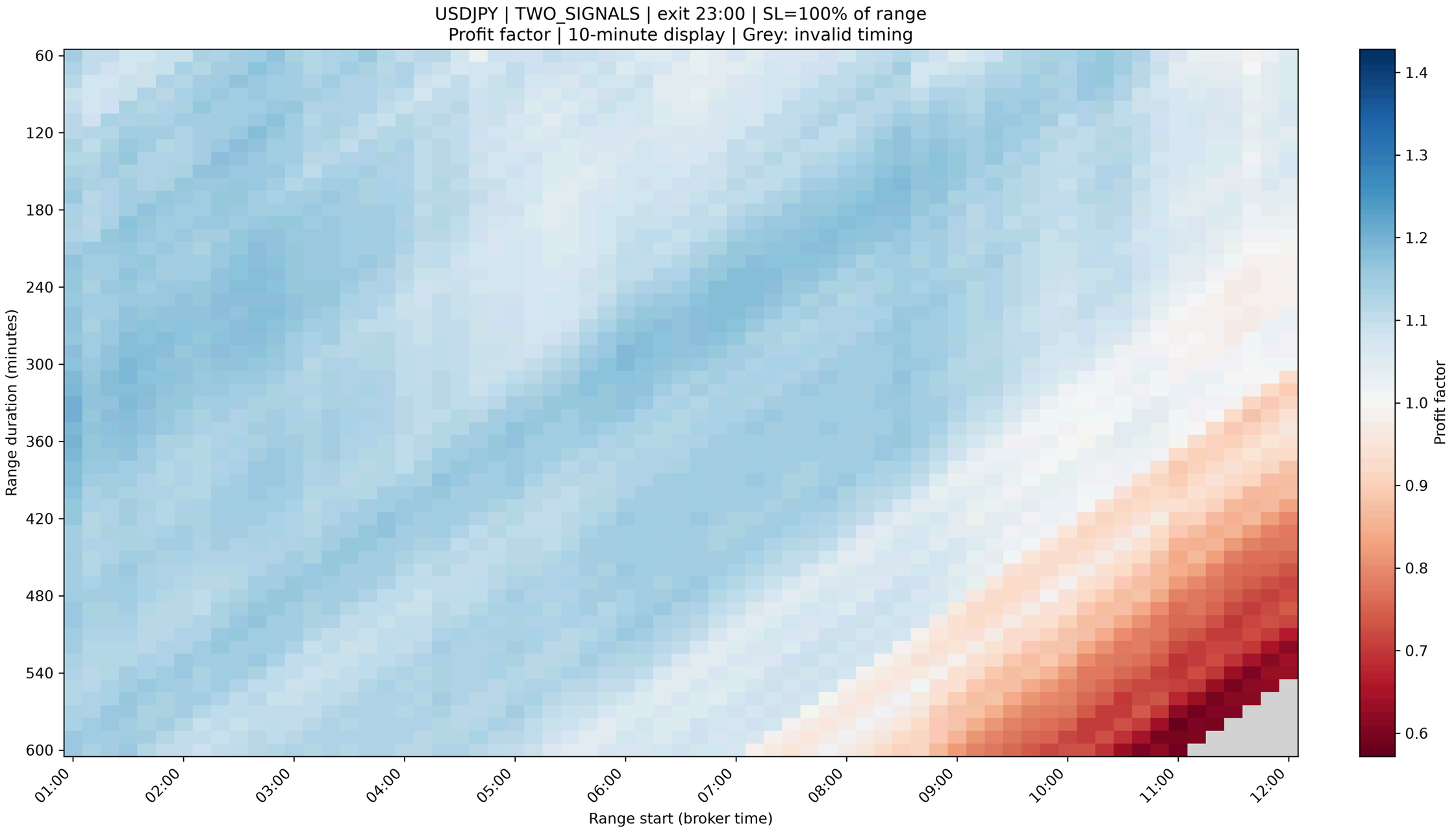
This measures consistency within the same historical sample. Independent validation remains outstanding.



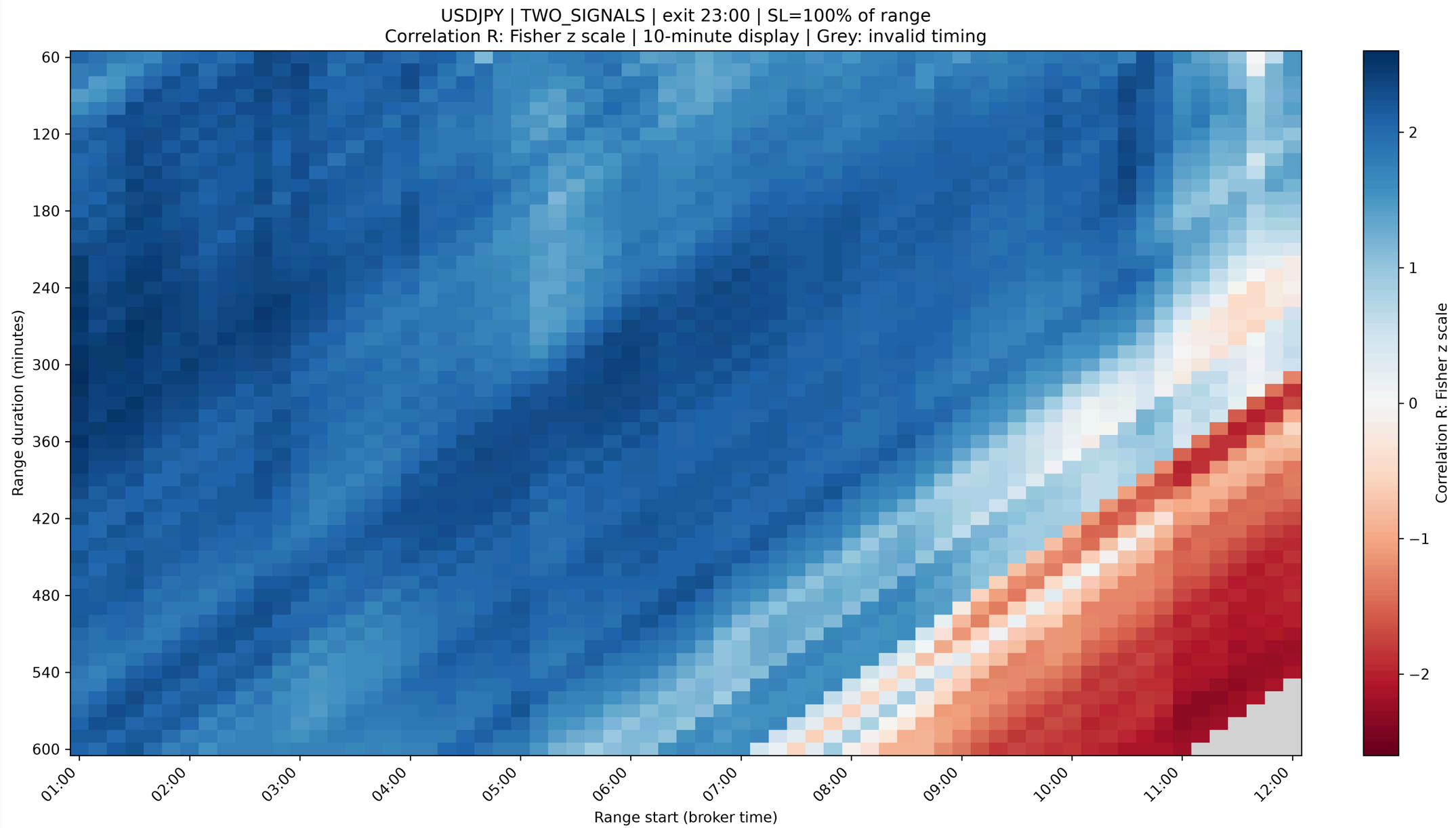
The five-minute grid contains 14,419 timing-valid settings for the 23:00 exit; the heatmap displays actual cells at ten-minute spacing. Early-starting ranges show broad historical profitability, while losses cluster among late-starting, long ranges. Grey cells indicate invalid timings. This finer grid explores the same historical sample as the original study.



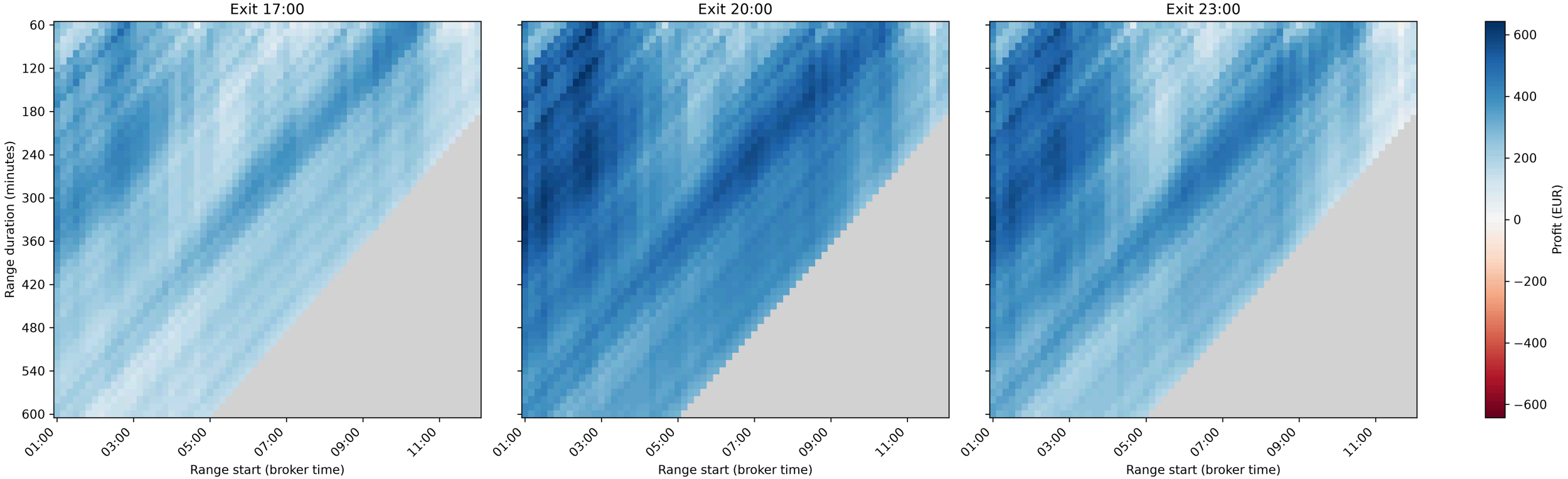
In the early-range region (starts 01:00–02:00; durations 270–390 minutes), all 325 five-minute settings were profitable. Recovery factor ranged from 3.0982 to 10.2059, with a median of 6.9081. The heatmap displays ten-minute spacing; statistics use the full five-minute grid. This region was selected after inspecting the historical results. Grey cells indicate invalid timings.



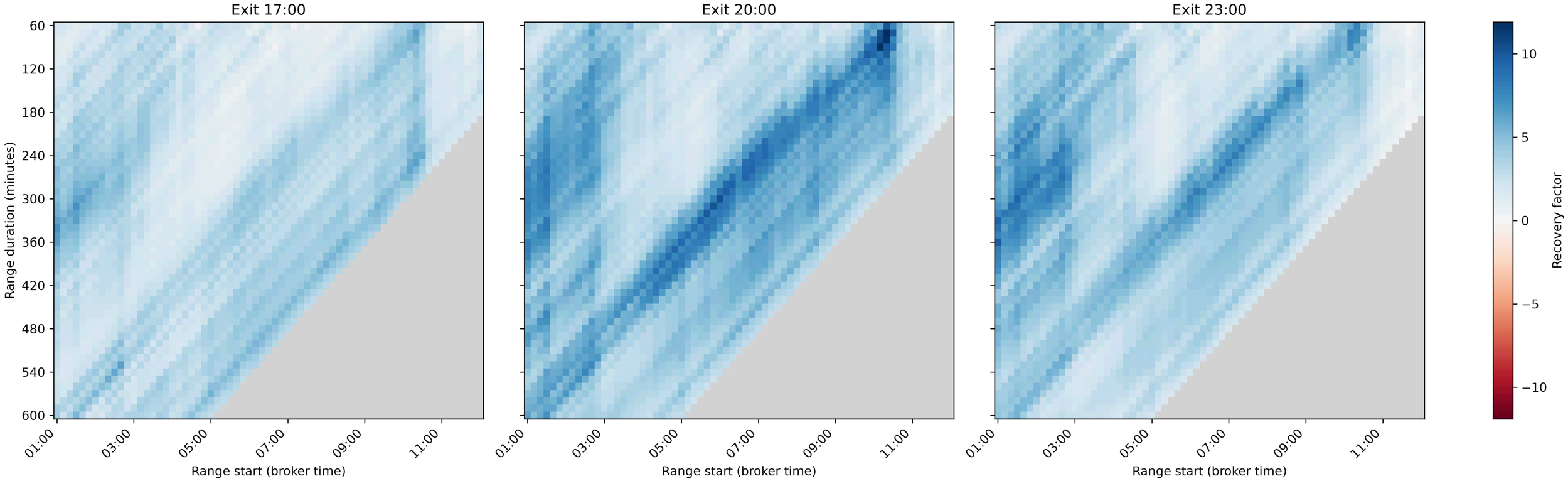
The neutral colour represents a profit factor of 1. In the early-range region (starts 01:00–02:00; durations 270–390 minutes), all 325 settings had profit factors above 1: median 1.1670, range 1.1150–1.2057. Statistics use the full five-minute grid; the heatmap displays ten-minute spacing. This region was selected after inspecting the historical results.



Fisher z(R) expands visual differences near ±1 while preserving the ordering of R values. In the 325-setting early-range region (starts 01:00–02:00; durations 270–390 minutes), raw R ranged from 0.9621 to 0.9891, with a median of 0.9832. Statistics use the full five-minute grid; the heatmap displays ten-minute spacing. This transformation changes the visual scale and does not provide independent validation.



Range starts and durations were tested in five-minute steps; heatmaps display ten-minute spacing. Median profit across all 10,927 settings valid for all three exits was €234.45 at 17:00, €406.39 at 20:00 and €323.22 at 23:00. All panels share one profit scale; grey cells lie outside the common comparison set. Results describe the same historical sample. Changing the exit also shifts the entry cutoff.



Across 10,927 settings valid for all three exits, 20:00 had the highest median profit, recovery factor and profit factor; 23:00 had the highest median R. Heatmaps display ten-minute spacing from five-minute scans and share one recovery-factor scale. Grey cells lie outside the common comparison set. These are exploratory results from the same historical sample.

Exit	Median profit	Median RF	Median PF	Median R
17:00	€234.45	3.0364	1.1453	0.9449
20:00	€406.39	4.4153	1.1774	0.9594
23:00	€323.22	3.7358	1.1283	0.9660

Findings and future work

Broader historical profitability among JPY pairs
Mean profitable-setting coverage was 54.15% for JPY pairs versus 5.16% for non-JPY pairs across the original 30-minute grid and both breakout modes. Performance varied substantially between pairs.
The difference extends beyond the two leading pairs
In two-signal mode, mean neighbourhood-supported coverage was 44.55% for JPY pairs versus 1.95% for non-JPY pairs. Removing USDJPY and CADJPY left a JPY mean of 27.53%.
USDJPY: regional consistency and exit-time sensitivity
Early-starting ranges showed consistent historical profitability across neighbouring settings. Across 10,927 matched five-minute settings, 20:00 had the highest median profit (€406.39), recovery factor (4.4153) and profit factor (1.1774) among the three fine-grid exits. The 23:00 exit had the highest median R (0.9660). These finer scans examine the same historical sample; the regions were identified after inspecting the results.
Future work
• Freeze a small shortlist and its selection rules.
• Compare representative configurations using real ticks and 1-minute OHLC.
• Check results with another broker’s data and trading conditions.
• Evaluate the frozen rules on genuinely unseen dates.
• Explore additional exit times, stop-loss/take-profit settings and account currency.
Interpretation
The evidence supports further investigation of JPY-linked breakouts. Direct tests of post-breakout price continuation and independent performance validation remain outstanding.